FGIS

Fundamentals of Geographic Information Systems

COMPLETE EXAM PREPARATION — ALL 5 UNITS

Most Important Questions | 2 Out of 4 Format | 2025–26

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HOW TO USE THIS DOCUMENT

This document covers the most important questions from all 5 units of FGIS based on past paper analysis. RED tags = appeared in both past papers (guaranteed). ORANGE = high probability. Since format is 2 out of 4, master the RED and ORANGE questions first.

Units covered: Unit I (GIS Basics) · Unit II (DBMS & GIS) · Unit III (GPS & Coordinates) · Unit IV (Spatial Analysis) · Unit V (Visualization & Maps)

UNIT 1 — GIS Basics & Spatial Data

Spatial Data Models · Topology · Vector vs Raster · TIN · Geographic Fields

Most Important Questions | Based on Past Paper Analysis | 2025-26

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q1. Define GIS. Explain its major capabilities and list five applications.

★ EXAM TIP: Asked in EVERY past paper — know this answer word for word.

Definition

A Geographic Information System (GIS) is a computer-based system designed to capture, store, manipulate, analyze, manage, and present spatially referenced geographic data. It integrates hardware, software, data, and people to support location-based decision-making.

Major Capabilities

- **Data Input/Capture:** Converts real-world data into digital form using GPS, remote sensing, digitizers.
- **Data Storage:** Stores spatial (geometry) and attribute (descriptive) data in an organized spatial database.
- **Data Manipulation:** Allows transformation, reprojection, editing, and correction of geographic data.
- **Spatial Query:** Enables location-based questions e.g. 'Find hospitals within 5 km of flood zone.'
- **Spatial Analysis:** Overlay, buffer, proximity, interpolation, and network analysis for decision-making.
- **Visualization:** Produces maps, 3D models, charts and reports to communicate geographic findings.

Five Applications of GIS

Application	Description
Urban Planning	Land-use zoning, infrastructure layout, city growth monitoring
Disaster Management	Flood zone mapping, evacuation route planning, risk assessment
Environmental Monitoring	Tracking deforestation, pollution spread, wildlife habitats
Transportation	Route optimisation, traffic analysis, logistics planning
Agriculture	Crop monitoring, soil analysis, precision irrigation using satellite data

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q2. Explain GIScience vs GISystems vs GIS Applications. Highlight key differences.

> ★ **EXAM TIP: Frequently asked as a short question. The table is your complete answer.**

Concept	Definition	Focus	Example
GIScience	Scientific study of concepts and theory underpinning GIS	Research & Theory	Spatial algorithms, projection theory
GISystems	Actual software and hardware tools to implement GIS	Technology	ArcGIS, QGIS, Google Earth Engine
GIS Applications	Practical use of GIS tools for real-world geographic problems	Real-world Use	Urban planning, disaster management

GIScience = WHY and HOW (theory). GISystems = WHAT (technology). GIS Applications = WHERE IT IS USED (practice).

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q3. Explain Topology in GIS. Describe connectivity, adjacency, and containment.

> ★ **EXAM TIP: Draw a simple diagram showing all 3 relationships — diagrams earn bonus marks.**

Definition of Topology

Topology is the mathematical study of spatial relationships and properties that remain unchanged under continuous deformation. In GIS it defines how features share geometry and relate to each other, ensuring data integrity and enabling accurate analysis.

Relationship	Definition	GIS Example
Connectivity	Two features share a common node or edge forming a network	Road segments meeting at an intersection
Adjacency	Two polygons share a common boundary — they are neighbours	Two states sharing a border

Relationship	Definition	GIS Example
Containment	One feature lies entirely within another feature	A lake inside a national park

- Prevents gaps and overlaps between polygons — ensures data integrity.
- Enables network analysis — road routing depends on connectivity.
- Supports spatial queries — find all features adjacent to a flood zone.

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q4. Compare Vector and Raster spatial data representation — characteristics, advantages, applications.

> ★ **EXAM TIP: Always include the comparison table. Mention 2 advantages and 2 use cases for each.**

Vector Data Model

Represents geographic features as precise geometric objects: points (single coordinate), lines (connected coordinates), and polygons (closed areas). Best for discrete, well-defined features.

Raster Data Model

Divides space into a regular grid of equal-sized cells (pixels). Each cell holds one attribute value. Best for continuous phenomena like elevation or temperature.

Characteristic	Vector	Raster
Representation	Points, Lines, Polygons	Regular grid of pixels/cells
Storage	Compact for discrete features	Large for high-resolution data
Accuracy	High — exact boundaries	Limited by cell size
Best For	Roads, boundaries, buildings	Elevation, satellite imagery, rainfall
Analysis	Network, overlay, proximity	Surface, terrain, interpolation
File Examples	Shapefile, GeoJSON, KML	GeoTIFF, DEM, satellite image

TIER 1 — APPEARED IN BOTH PAST PAPERS

Q5. Explain concept of Tessellation and its forms. Also explain TIN.

Tessellation

Tessellation is the process of partitioning geographic space into non-overlapping, space-filling tiles with no gaps between them. It is the fundamental concept behind raster GIS data representation.

Type	Description	Example
Regular	Uniform equal-sized shapes — squares, triangles, hexagons	Raster grid — satellite image pixels
Irregular	Variable-sized polygons based on data distribution	Voronoi polygons, TIN triangles
Hierarchical	Progressive subdivision into smaller units at finer levels	Quadtree — each cell splits into 4 sub-cells

Triangulated Irregular Network (TIN)

A TIN is a vector-based 3D terrain surface model created by connecting irregularly distributed elevation points into non-overlapping triangles using Delaunay triangulation. Each triangle represents a planar surface with constant slope and aspect.

- More points placed in complex terrain — adapts to surface variation.
- Derived products: slope maps, aspect maps, contour lines, volume calculations.
- Advantage over raster DEM: smaller file size, adaptive detail where needed.

TIER 1 — HIGH PROBABILITY

Q6. Define a Geographic Field. Explain its data types and value types with examples.

A geographic field assigns a value to every location in a study area. It represents phenomena that vary continuously across space — every point has a defined value (e.g. temperature, elevation, air pressure).

Data Type	Description	Example
Nominal	Named categories, no order	Soil type: clay, sand, loam
Ordinal	Ordered categories, no numeric difference	Flood risk: low, medium, high
Interval	Numeric, equal intervals, no true zero	Temperature in Celsius
Ratio	Numeric with true zero — most precise	Elevation in metres above sea level

Value Type	Description	Example
Continuous	Values change gradually across space	Elevation, temperature, humidity
Categorical	Discrete class values assigned to regions	Vegetation type, land use class
Binary	Only two values: 0 or 1 (True/False)	Presence or absence of a species

TIER 1 — HIGH PROBABILITY

Q7. Explain Spatio-Temporal Data Models. Describe notions of time used in GIS.

Spatio-temporal data models represent geographic phenomena that have both a spatial location AND change over time. They combine WHERE with WHEN to track how features evolve.

Time Concept	Description	GIS Example
Continuous	Time flows without interruption — values exist at every instant	Temperature recorded every second by a weather sensor
Discrete	Data captured only at specific snapshot moments	Population census every 10 years
Cyclical	Time follows a regularly repeating pattern	Annual monsoon flooding; seasonal crop growth
Ordered	Events in a defined sequence — each stage follows the previous	Forest → Farmland → Urban settlement

- Enable change detection — comparing two time periods to find what changed and where.
- Support forecasting — predict future urban growth, flood extents, climate shifts.
- Essential for disaster monitoring, environmental management, and climate studies.

UNIT 2 — DBMS & GIS

Relational Model · Spatial Databases · SDI · Functional Components

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TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q1. List and explain the functional components of GIS. Explain any two in detail.

> ★ **EXAM TIP: Asked in both papers as 'List functional components' and 'Explain any two in detail.'**

Five Functional Components of GIS

Component	Role	Example
Hardware	Physical devices for input, processing, storage and output	GPS receiver, server, scanner, plotter
Software	Programs for data management, analysis and display	ArcGIS, QGIS, MapInfo, PostGIS
Data	Spatial + attribute data — the most critical component	Satellite images, shapefiles, DEMs, census data
People	Users who design, operate, analyse and manage the GIS	GIS analysts, urban planners, scientists
Methods	Procedures and workflows applied in spatial analysis	Overlay analysis, buffering, network routing

Component 1 — Data (in detail)

Data is the most important component of GIS — without accurate, complete data, no analysis is meaningful. GIS data has two parts: (1) Spatial data: coordinates and geometry defining location and shape. (2) Attribute data: descriptive information about each feature. Both are stored together in a spatial database linked by a unique Feature ID.

Component 2 — Software (in detail)

GIS software provides the tools to input, manage, query, analyse and visualize spatial data. Modern GIS software supports: data format conversion, coordinate reprojection, spatial query with SQL, overlay and buffer operations, map rendering and export. Examples: ArcGIS (commercial, industry standard), QGIS (open-source), Google Earth Engine (cloud-based raster analysis).

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q2. Explain the Relational Data Model. Describe relations, tuples, and attributes.

> ★ **EXAM TIP: Both papers asked about relational model. The table format with examples is perfect for this answer.**

Relational Data Model

The relational data model organizes data into tables (relations) consisting of rows (tuples) and columns (attributes). Each table represents one entity type. Tables are linked by shared key fields — enabling complex queries without data duplication. It was introduced by E.F. Codd in 1970 and is the foundation of all modern GIS databases.

Term	Definition	GIS Example
Relation (Table)	A 2D table storing data about one entity type	ROADS table — stores all road features
Tuple (Row)	One record/instance in the table — one real-world feature	One row = one road segment
Attribute (Column)	A specific property or field for every record	RoadName, Length_km, SurfaceType, SpeedLimit
Primary Key	Unique identifier for each tuple — no duplicates	ROAD_ID: 1001, 1002, 1003...
Foreign Key	Links one table to another table's primary key	DISTRICT_ID in ROADS links to DISTRICT table

Why Relational Model is Used in GIS

- Eliminates data redundancy — store district name once, reference by ID everywhere.
- Supports SQL queries — SELECT, WHERE, JOIN operations for spatial analysis.
- Enables relationships between tables — roads linked to districts, parcels linked to owners.
- ACID compliance — ensures data consistency and integrity in multi-user environments.

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q3. Explain how GIS is integrated with DBMS. What are the benefits?

GIS-DBMS Integration

Traditional GIS stored spatial data in flat files (shapefiles) separate from attribute data. Integration with DBMS (Database Management Systems) combines both in one environment — spatial data stored as geometry objects in the same database tables as attribute data. Technologies like PostGIS (PostgreSQL extension) and Oracle Spatial enable this integration.

Benefits of GIS-DBMS Integration

Benefit	Description
Unified Storage	Spatial geometry and attributes stored together — one query retrieves both
Spatial SQL	Use standard SQL extended with spatial functions: ST_Distance(), ST_Intersects(), ST_Buffer()
Multi-user Access	Multiple users query and edit data simultaneously — with transaction locking

Benefit	Description
Data Integrity	Constraints, triggers and topology rules enforce data quality automatically
Scalability	Handle millions of features efficiently using spatial indexes (R-tree, quad-tree)
Interoperability	Standard formats (WKT, WKB) allow data exchange between different GIS systems

TIER 1 — HIGH PROBABILITY

Q4. Write a short note on Spatial Data Infrastructure (SDI). Explain its importance.

What is SDI?

Spatial Data Infrastructure (SDI) is a framework of policies, standards, technologies, data, and human resources that enables the sharing, access, and use of geographic data across organizations, agencies, and countries. It ensures that spatial data collected by one organization can be discovered and used by others without duplication.

Key Components of SDI

- **Data:**The actual spatial datasets — topographic, administrative, environmental, demographic.
- **Metadata:**Catalogue describing each dataset's content, source, accuracy, and format.
- **Standards:**Common data formats and protocols (OGC, ISO 19115) enabling interoperability.
- **Clearinghouse/Portal:**Online repository where datasets can be discovered and downloaded.
- **Policies:**Legal frameworks governing data ownership, sharing rights, and privacy.

Importance of SDI

- Eliminates duplication — organizations share data instead of each collecting it independently.
- Supports national planning — government agencies access consistent, unified spatial data.
- Enables disaster response — multiple agencies access real-time data from one platform.
- Examples: India's NSDI (National Spatial Data Infrastructure), ISRO Bhuvan portal, EU INSPIRE directive.

TIER 1 — HIGH PROBABILITY

Q5. Discuss advantages of using DBMS instead of file-based systems for GIS data management.

Aspect	File-Based System	DBMS
Data redundancy	Same data stored in multiple files — duplicated	Centralized — stored once, referenced many times

Aspect	File-Based System	DBMS
Data integrity	No automatic validation — errors propagate	Constraints and triggers enforce rules automatically
Multi-user access	File locking — one user at a time	Concurrent access with transaction management
Querying	Manual search through files	Powerful SQL queries — SELECT, JOIN, WHERE
Security	File-level permissions only	Role-based access control per table/column
Scalability	Slow for large datasets	Spatial indexes handle millions of features efficiently
Backup & Recovery	Manual — prone to data loss	Automated backup with point-in-time recovery

TIER 1 — HIGH PROBABILITY

Q6. Explain what are pixels and voxels. Explain database relationships.

Pixels and Voxels

Term	Definition	Dimensions	GIS Application
Pixel	Smallest unit of a 2D raster image — each pixel stores one attribute value	2D (x, y)	Satellite imagery, aerial photos, DEMs
Voxel	3D equivalent of a pixel — smallest unit in a 3D volumetric grid — stores a value for a 3D location	3D (x, y, z)	Underground geology, 3D city models, atmospheric data

Database Relationships

Relationship	Description	GIS Example
One-to-One (1:1)	One record in Table A links to exactly one record in Table B	One road segment has one detailed maintenance record
One-to-Many (1:N)	One record in Table A links to many records in Table B	One district contains many road segments
Many-to-Many (M:N)	Many records in A link to many records in B — requires a junction table	Many parcels can have many owners; many owners can own many parcels

TIER 1 — HIGH PROBABILITY

Q7. Explain database indexing and its importance in GIS data retrieval.

What is Database Indexing?

A database index is a data structure that improves the speed of data retrieval operations on a table by allowing the database engine to find rows without scanning the entire table. In GIS, spatial indexing enables fast location-based queries over millions of features.

Types of Spatial Indexes

Index Type	Description	Use in GIS
R-tree (Rectangle tree)	Organizes features by their minimum bounding rectangles in a hierarchical tree	Most common spatial index — used by PostGIS, Oracle Spatial, Esri geodatabases
Quad-tree	Recursively divides space into four quadrants based on feature distribution	Used in raster data storage and point cloud data
Grid index	Divides space into a regular grid — features indexed by which cell they fall in	Efficient for uniformly distributed features

- Without indexing: database scans all features one by one — very slow for large datasets.
- With indexing: database jumps directly to relevant features — queries are 100x-1000x faster.
- Spatial indexes are essential for real-time web map applications serving millions of users.

UNIT 3 — GPS, Coordinates & Data Quality

Map Projections · GPS · Geoid/Ellipsoid · Coordinate Systems · Data Quality

Most Important Questions | Based on Past Paper Analysis | 2025-26

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q1. What is a Map Projection? Explain major classifications with examples.

> **★ EXAM TIP: Asked in both papers. Know the three classification systems and draw the projection surface diagram.**

What is a Map Projection?

A map projection is a mathematical method used to transform the curved 3D surface of the Earth onto a flat 2D map surface. Since the Earth is a sphere/ellipsoid, it is geometrically impossible to represent it on a flat surface without some distortion. Every projection distorts at least one of: shape, area, distance, or direction.

Classification by Distortion Property (Preserved)

Type	Property Preserved	Distortion	Example
Conformal (Orthomorphic)	Shape of small areas	Area distorted	Mercator Projection
Equal Area (Equivalent)	Area relationships	Shape distorted	Albers Equal Area
Equidistant	Distance from one or two points	Both shape and area distorted	Azimuthal Equidistant
Azimuthal	Direction from center point	Shape and area at edges	Lambert Azimuthal

Classification by Projection Surface

Surface	Description	Example
Cylindrical	Imaginary cylinder wrapped around Earth — then unrolled flat	Mercator — good for navigation
Conical	Imaginary cone placed over Earth — then unrolled	Albers — good for mid-latitude countries
Planar (Azimuthal)	Flat plane touches Earth at one point	Stereographic — good for polar regions

Why Map Projections are Needed in GIS

- All GIS displays are flat screens or printed maps — 3D data must be flattened.
- Different projections minimize different types of distortion for different purposes.
- Consistent projection across layers is essential — mismatched projections cause spatial errors.

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q2. Explain the working principle of GPS. Describe how satellites, signals, and receivers work together.

> ★ **EXAM TIP: Both papers asked about GPS/NAVSTAR. Know the 3 segments and trilateration principle clearly.**

GPS — Three Segments

Segment	Components	Role
Space Segment	24+ GPS satellites in 6 orbital planes at ~20,200 km altitude	Continuously broadcast position and time signals on two frequencies (L1, L2)
Control Segment	Master Control Station (Colorado) + ground monitoring stations worldwide	Monitor satellite health, update orbital data, synchronize atomic clocks
User Segment	GPS receivers in phones, vehicles, survey equipment, drones	Receive signals from multiple satellites, calculate position using trilateration

How GPS Determines Position — Trilateration

- **Step 1:**Receiver captures signals from at least 4 satellites simultaneously.
- **Step 2:**Receiver calculates distance to each satellite using: Distance = Speed of light × Signal travel time.
- **Step 3:**With 3 satellites — position is narrowed to a point in 3D space (latitude, longitude, elevation).
- **Step 4:**4th satellite is needed to correct receiver clock errors and achieve accurate positioning.

Common Sources of GPS Errors

Error Type	Cause	Impact
Atmospheric delay	Ionosphere and troposphere slow down signals	Position error of 1-10 metres
Multipath error	Signals bounce off buildings before reaching receiver	Urban canyon effect — inaccurate positioning
Satellite geometry (GDOP)	Poor satellite spread across the sky	Dilutes precision — wide GDOP = less accurate
Clock errors	Satellite and receiver clock imperfections	Timing errors cause distance calculation errors
Ephemeris errors	Inaccurate satellite orbital data broadcast	Small but systematic positional errors

TIER 1 — HIGH PROBABILITY

Q3. Write a short note on Geoid and Ellipsoid. Explain their importance in geographic measurements.

The Geoid

The geoid is an irregular-shaped model of the Earth that represents the equipotential surface of Earth's gravity field — it corresponds to mean sea level extended continuously under all land areas. The geoid is lumpy and irregular because Earth's mass is not uniformly distributed. GPS heights are measured relative to the ellipsoid and then converted to geoid heights (orthometric heights) for practical use.

The Ellipsoid (Spheroid)

The ellipsoid is a smooth mathematical approximation of Earth's shape — slightly flattened at the poles and bulging at the equator. It is defined by two parameters: semi-major axis (a) at the equator and semi-minor axis (b) at the poles. The ellipsoid is used as the mathematical surface for coordinate systems and map projections.

Model	Shape	Used For	Example
Geoid	Irregular — follows gravity equipotential	Elevation/height measurement — orthometric heights	Mean sea level reference for surveying
Ellipsoid	Smooth mathematical sphere — slightly flattened	Map projections, GPS coordinates, datum definition	WGS84 (used by GPS), GRS80

- WGS84 (World Geodetic System 1984) is the standard ellipsoid used by GPS worldwide.
- The difference between ellipsoid height and geoid height is called the geoid undulation (N).

TIER 1 — HIGH PROBABILITY

Q4. Explain GPS, GLONASS, and Galileo positioning technologies.

System	Full Name	Country/Agency	Satellites	Coverage	Accuracy
GPS	Global Positioning System	USA (DoD)	31 active satellites	Global	~3-5m civil; <1m military
GLONASS	Globalnaya Navigatsionnaya Sputnikovaya Sistema	Russia	24 satellites	Global	~3-5m civil
Galileo	Galileo Navigation Satellite System	European Union (ESA)	30 planned satellites	Global	~1m civil (best civilian accuracy)
NavIC	Navigation with Indian Constellation	India (ISRO)	7 satellites	India + 1500 km region	~5m standard; ~0.5m restricted

- Using multiple systems (GPS + GLONASS + Galileo) simultaneously improves accuracy and reliability.
- Modern smartphones receive all three systems simultaneously for best positioning.

TIER 1 — HIGH PROBABILITY

Q5. Explain what is interpolation in GIS. How is it used to generate continuous surfaces?

What is Interpolation?

Interpolation is the process of estimating values at unsampled locations based on known values at surrounding sampled points. In GIS, it is used to create continuous surface models (raster grids) from a set of discrete point measurements — for example, predicting rainfall across an entire region from a sparse network of rain gauges.

Common Interpolation Methods

Method	Description	Best For
Inverse Distance Weighting (IDW)	Closer points have more influence than distant ones — weight = 1/distance	Smooth, gradual surfaces — temperature, rainfall
Kriging	Statistical method using spatial autocorrelation — accounts for spatial trends and variability	Most accurate for scientific data — soil properties, groundwater
Spline	Fits a mathematically smooth curve through all data points	Terrain surfaces, contour generation
Natural Neighbour	Uses Voronoi polygons to determine influence of surrounding points	Elevation data from irregularly spaced surveys
Trend Surface	Fits a polynomial equation to the overall spatial trend	Regional-scale phenomena — regional temperature trends

TIER 1 — HIGH PROBABILITY

Q6. Explain the concept of data quality in GIS. Discuss the five dimensions.

What is Data Quality in GIS?

Data quality refers to how accurately and completely a spatial dataset represents the real-world phenomena it was designed to model. High data quality means the data is accurate, consistent, current, and complete — enabling reliable analysis and trustworthy decision-making.

Dimension	Definition	Example
Positional Accuracy	How correctly features are located relative to their true ground position	GPS road survey accurate to ±2m vs. digitized map accurate to ±50m

Dimension	Definition	Example
Attribute Accuracy	How correct the descriptive values associated with features are	Land use classified correctly as 'residential' not misclassified as 'commercial'
Temporal Accuracy	Whether the data represents the correct time period and is sufficiently current	2024 road network data used for current navigation — not a 2010 dataset
Completeness	Whether all required features and attributes are present in the dataset	All roads in a district are mapped — no missing segments or villages
Logical Consistency	Whether spatial relationships are internally valid — no gaps, overlaps, or topology errors	No overlapping land parcels; all road segments properly connected at intersections

TIER 1 — HIGH PROBABILITY

Q7. Explain coordinate transformation in GIS. Why is it required?

What is Coordinate Transformation?

Coordinate transformation is the mathematical process of converting spatial data from one coordinate reference system (CRS) to another. It is required whenever spatial data from different sources use different projections or datums — without transformation, features from different layers will not align correctly.

Types of Coordinate Transformation

Type	Description	When Used
Geographic Transformation (Datum Shift)	Converts between two different datums — e.g., local datum to WGS84	When combining old survey data with modern GPS data
Projection Transformation	Converts between two different map projections — e.g., UTM to Geographic	When combining layers in different projected CRS
Affine Transformation	Linear transformation — translation, rotation, scaling, shearing	Georeferencing scanned maps to fit real-world coordinates

- On-the-fly reprojection: modern GIS software like QGIS and ArcGIS can reproject data automatically during display — but analysis should always use data in a consistent CRS.
- Datum transformation is critical — using wrong datum can introduce errors of 100-200 metres.

UNIT 4 — Spatial Analysis

Overlay · Network Analysis · Neighbourhood Functions · Error Propagation · Terrain

Most Important Questions | Based on Past Paper Analysis | 2025-26

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q1. What are the analytical capabilities of GIS? Explain retrieval, classification, and measurement.

> **★ EXAM TIP: Both papers asked analytical capabilities. Know all four categories plus the three functions.**

Four Categories of GIS Analytical Capabilities

Category	Description	Example
Retrieval & Query	Select and retrieve spatial features based on location or attribute criteria	Find all schools within 2 km of a river
Measurement	Calculate geometric properties of features — distance, area, perimeter, volume	Calculate total area of forest cover in a district
Classification	Group features into categories based on attribute values or spatial position	Reclassify elevation raster into: low / medium / high zones
Overlay & Analysis	Combine multiple layers to derive new spatial information and relationships	Overlay soil + slope + rainfall to find optimal farming land

Retrieval Functions

Retrieval extracts features from the database that meet specified conditions using Spatial Selection Queries: (1) Point-in-polygon: which zone does this location fall in? (2) Region query: find all features within a specified area. (3) Attribute query: SELECT * FROM roads WHERE surface_type = 'Asphalt'.

Classification Functions

Classification groups continuous data into discrete categories: Equal interval (divides value range into equal parts), Quantile (equal number of features per class), Natural breaks/Jenks (maximizes differences between classes), Standard deviation (classes based on statistical spread from mean).

Measurement Functions

- Distance measurement: Euclidean (straight-line), Manhattan (grid), Geodesic (along curved Earth surface).
- Area calculation: computed from polygon geometry in square metres, hectares, or km².
- Perimeter/length: total boundary length of a polygon or length of a line feature.
- Volume: computed from raster DEMs — e.g., volume of a reservoir at a given water level.

Q2. What is Vector Overlay in GIS? Explain types. Differentiate Vector vs Raster Overlay.

> ★ **EXAM TIP: Both papers asked overlay. Know all vector overlay types and be able to draw a diagram.**

Vector Overlay

Vector overlay combines two or more vector polygon layers to produce a new layer containing the combined geometry and attributes of both input layers. The boundaries of all input layers are merged and new polygons are formed at every intersection.

Overlay Type	Description	Output	Example
Union	Combines ALL features from BOTH layers — nothing is discarded	All areas from A and B	Combine two district layers — keep all districts
Intersect	Keeps only the areas where both layers OVERLAP — rest discarded	Only overlapping area	Find areas that are both forest AND protected reserve
Clip	Cuts input layer to the shape of a clip mask layer	Input features within clip boundary	Cut road network to just Mumbai city boundary
Erase (Difference)	Removes from input layer the areas that overlap with erase layer	Input minus erase area	Remove flood zones from available land parcels
Identity	Preserves all of input layer + adds attributes from identity layer where they overlap	Full input + identity attributes added	Add district names to road segments that pass through each district

Raster Overlay

Raster overlay combines two or more raster grids cell by cell using mathematical or logical operations (map algebra). Each output cell value is computed from the corresponding cells in the input rasters.

Aspect	Vector Overlay	Raster Overlay
Method	Geometric intersection of polygon boundaries	Cell-by-cell mathematical operations
Speed	Slower — complex geometry calculations	Fast — simple cell-by-cell arithmetic
Output	New polygons with merged attributes	New raster grid with computed cell values
Best for	Precise boundary operations, land parcels	Continuous surface analysis, suitability modelling

Q3. What are Neighbourhood Functions in GIS? Explain any four types.

What are Neighbourhood Functions?

Neighbourhood functions analyze each location in a GIS dataset based on the characteristics of its surrounding area (neighbourhood). They compute a new output value for each cell or feature based on the values of nearby features within a defined neighbourhood window or distance.

Function	Description	Application
Buffering (Proximity)	Creates a zone of specified distance around a point, line, or polygon feature	Find all properties within 500m of a proposed highway
Focal Statistics	Calculates statistics (mean, max, min, sum) for each raster cell based on values in surrounding cells within a moving window	Smoothing a noisy elevation surface using a 3x3 mean filter
Thiessen/Voronoi Polygons	Divides space so each polygon contains the area closest to its seed point — any location in a polygon is nearer to its seed than to any other seed	Nearest hospital analysis — which hospital serves each residential area
Slope & Aspect	Computes the rate of elevation change (slope) and the direction of steepest descent (aspect) from a DEM for each cell	Identify landslide risk zones; determine solar panel orientation
Viewshed Analysis	Determines which areas can be seen from a given viewpoint on a terrain surface	Locate cell towers; determine visibility from a watchtower
Flow Direction & Accumulation	Determines direction water flows across terrain and accumulates flow from upslope areas	Watershed delineation, flood routing, stream network extraction

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q4. What is Network Analysis in GIS? Describe common applications.

What is Network Analysis?

Network analysis in GIS performs analytical operations on connected linear networks (roads, pipelines, rivers, utility lines). It models flows, routes, and connectivity through a system of nodes (junctions) and edges (line segments) with associated costs (distance, time, capacity).

Common Network Analysis Operations

Operation	Description	Real-World Application
Shortest Path	Finds the route between two points that minimizes cost (distance, time, fuel)	GPS navigation routing; delivery optimisation

Operation	Description	Real-World Application
Closest Facility	Finds the nearest facility (from a set) to each demand location	Find nearest hospital, fire station, or pharmacy to an emergency
Service Area	Determines the area reachable from a facility within a given cost threshold	Find all homes within 10-minute drive of a school
Vehicle Routing	Optimises routes for multiple vehicles serving multiple stops — minimizes total cost	Logistics — delivery trucks visiting many locations efficiently
Network Flow	Models flow capacity and bottlenecks in a network	Water distribution network, electrical grid capacity, traffic flow
Travelling Salesman	Finds the optimal order to visit multiple stops returning to start	Sales representative visiting multiple client locations

TIER 1 — HIGH PROBABILITY

Q5. Explain proximity analysis and flow computation in GIS.

Proximity Analysis

Proximity analysis measures and evaluates the spatial relationships between features based on their distance from each other. It answers questions about nearness, accessibility, and spatial separation.

- **Buffer zones:**Create distance-based zones around features — 500m buffer around schools identifies restricted construction areas.
- **Point distance:**Calculate distance from every feature to its nearest neighbour — identifies isolated or clustered features.
- **Cost distance:**Measures distance weighted by travel cost (friction surface) — accounts for terrain, road type, speed limits.
- **Allocation:**Assigns each location to its nearest service centre — used for school zone delineation, hospital catchments.

Flow Computation

Flow computation in GIS models how substances (water, traffic, utilities) move through a network or across a terrain surface. Based on terrain analysis using DEMs.

- **Flow direction:**Determines which of 8 neighbouring cells each cell drains into — based on elevation differences.
- **Flow accumulation:**Counts the number of upslope cells that drain into each cell — high accumulation = streams and rivers.
- **Stream network extraction:**Cells with accumulation above a threshold are classified as streams.
- **Watershed delineation:**Identifies the catchment area that drains to a specific outlet point — critical for flood modelling.

TIER 1 — HIGH PROBABILITY

Q6. What is Error Propagation in GIS? How can it be measured and controlled?

What is Error Propagation?

Error propagation in GIS refers to the way that errors in input spatial data spread, accumulate, and magnify as data passes through spatial operations. When multiple data layers — each with their own positional and attribute errors — are combined in overlay or analysis operations, the errors in each layer compound into potentially larger errors in the output.

Sources of Errors

- **Positional error:**Features are not located at their true coordinates — from GPS inaccuracy, digitizing, or scan distortion.
- **Attribute error:**Wrong values assigned to features — misclassification during data collection.
- **Processing error:**Rounding during coordinate calculations, generalisation during scale changes.
- **Model error:**Simplifications in the analysis model that don't fully represent reality.

How to Measure and Control Error Propagation

Method	Description
Error budgeting	Track the known accuracy of each input layer and estimate maximum possible output error
Monte Carlo simulation	Run analysis thousands of times with small random input variations — measure output variability
Sensitivity analysis	Test how much output changes when each input layer error is varied slightly
Data quality standards	Use only data meeting minimum accuracy standards (ISO 19157) for critical analysis
Topology validation	Check for and fix topological errors before running overlay operations
Metadata documentation	Document error sources in output metadata so future users understand data limitations

TIER 1 — HIGH PROBABILITY

Q7. What is Terrain Modelling in GIS? Explain raster-based surface analysis.

Terrain Modelling

Terrain modelling in GIS is the digital representation and analysis of the Earth's surface topography using spatial data. It enables quantitative analysis of landscape features — slopes, valleys, ridges, drainage patterns — that are essential for hydrology, engineering, environmental planning, and hazard assessment.

Raster-Based Surface Analysis Products

Analysis	Description	Application
Slope	Rate of elevation change — expressed in degrees or percent	Landslide risk, road gradient assessment, farm erosion potential

Analysis	Description	Application
Aspect	Direction of steepest descent from each cell — compass direction (0-360°)	Solar energy potential, snow melt modelling, vegetation analysis
Hillshade	Simulates sunlight illumination for visual 3D impression of terrain	Background for topographic maps, visual terrain communication
Contour generation	Extract lines of equal elevation from DEM at regular intervals	Topographic maps, engineering design, flood level mapping
Curvature	Measures convexity/concavity of the surface — rate of slope change	Soil erosion modelling, flow concentration analysis
Viewshed	Determines visible areas from a given viewpoint on the terrain	Cell tower placement, military line-of-sight, tourism vista points

UNIT 5 — Visualization & Cartography

Bertin's Variables · Terrain Visualization · Map Design · Dissemination · Temporal Maps

Most Important Questions | Based on Past Paper Analysis | 2025-26

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q1. List and explain Bertin's six Visual Variables used in map design.

> ★ **EXAM TIP: Appeared in BOTH past papers. Memorise: SSCOOT — Size, Shape, Colour hue, cOlour value, Orientation, Texture.**

What are Visual Variables?

Visual variables are the graphic properties of map symbols that can be varied to represent different types of geographic information. Defined by French cartographer Jacques Bertin in 1967, they are the fundamental building blocks of all map symbolisation.

Visual Variable	Definition	Best Data Type	Map Example
Size	Variation in area or length of a symbol — larger = greater value	Quantitative — shows magnitude	Proportional circles — larger circle = higher population
Shape	Variation in geometric form — circle, square, triangle, star	Nominal — distinguishes categories	Hospital = cross, School = star, Police = shield
Colour Hue	Variation in actual colour — red, blue, green, yellow	Nominal — distinguishes categories	Land use: green = forest, blue = water, red = urban
Colour Value	Variation in lightness to darkness of a colour	Ordinal & Quantitative — shows intensity	Choropleth: light yellow to dark red = increasing density
Orientation	Variation in angle or direction of a symbol — 0°, 45°, 90°	Ordinal/Qualitative — shows direction	Wind arrows; geology hatching at different angles
Texture / Grain	Variation in coarseness or pattern spacing of fill	Qualitative & Ordinal — differentiates areas	Fine dots vs coarse dots vs diagonal lines for land cover

Data Type	Best Visual Variables
Nominal (categories — no order)	Colour Hue, Shape
Ordinal (ranked — low/medium/high)	Colour Value (light to dark), Size, Texture
Quantitative (numeric values)	Size (proportional), Colour Value

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q2. Explain the Visualization Process in GIS. What is 'How do I Say What to Whom'?

> ★ **EXAM TIP: This exact phrase was a direct question in Paper 1. The table for 'Say What to Whom' is essential.**

Visualization Process in GIS

- **Data Collection:** Raw spatial data gathered from GPS, sensors, satellites, existing databases.
- **Data Processing:** Data cleaned, transformed, projected, and classified into meaningful categories.
- **Symbolisation:** Features assigned visual symbols using Bertin's visual variables.
- **Map Design:** Cartographic elements added — title, legend, scale bar, north arrow, graticule.
- **Output & Dissemination:** Final map exported as print, PDF, web map, or interactive digital format.

'How Do I Say What to Whom — and Is It Effective?'

Question	Meaning	Cartographic Decision
HOW?	What visual form should convey the information?	Choose map type: choropleth, proportional symbol, isoline, flow map
WHAT?	What geographic message or story is being communicated?	Select data, classification method, and level of generalisation
To WHOM?	Who is the audience — experts, public, children, policymakers?	Adjust complexity, colour scheme, vocabulary, level of detail
EFFECTIVE?	Does the map successfully communicate without confusion?	Evaluate using clarity, accuracy, visual hierarchy, user feedback

Exploring vs Presenting Data

Aspect	Exploring Data	Presenting Information
Purpose	Discover patterns and anomalies in data	Communicate a specific finding to an audience
Audience	The analyst — internal use	External stakeholders — public, decision-makers
Map style	Rough, flexible, unpolished	Polished, carefully designed, publication-ready
Example	Analyst zooming into flood data finding risk areas	Final flood risk map published in a government report

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q3. Explain Map Cosmetics and their role in improving map readability.

> ★ **EXAM TIP: Also called 'cartographic toolbox'. List all elements and explain WHY each matters.**

What are Map Cosmetics?

Map cosmetics (map furniture/cartographic elements) are the non-data components added to make a map self-explanatory, professional, and easy to read. A map without cosmetics is incomplete and potentially misleading.

Element	Description	Why It Matters
Title	Descriptive name stating what the map shows	Tells the reader immediately what they are looking at
Legend (Key)	Explains what each symbol, colour and pattern represents	Without legend the map cannot be correctly interpreted
Scale Bar	Graphical ratio of map distance to ground distance	Allows measurement of real distances directly on the map
North Arrow	Indicates direction of geographic north	Orients the reader and establishes directional context
Graticule / Grid	Network of latitude/longitude lines or coordinate grid	Enables precise geographic location referencing
Border / Neatline	Frame defining the map boundary	Clean, professional boundary — defines the map extent
Data Source & Date	Credits data provider and collection date	Enables reader to assess data reliability and currency
Inset Map	Small overview map showing wider geographic context	Helps unfamiliar readers locate the study area on a larger map
Colour Scheme	Consistent, purposeful colour choices aligned with data type	Improves readability; supports correct interpretation

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q4. Describe methods to represent terrain elevation: Contour maps, DEM, and Hillshading.

Method 1 — Contour Lines

Lines connecting points of equal elevation. Closely spaced = steep slope. Widely spaced = gentle terrain. Closed contour with higher values inside = hill. Contour interval = fixed elevation difference between adjacent lines (e.g. every 20m).

Method 2 — Digital Elevation Model (DEM)

Raster grid where each cell stores the terrain elevation in metres. Sources: SRTM (30m global free data), LiDAR (centimetre accuracy), photogrammetry, GPS surveys. Derived products: slope, aspect, hillshade, watershed, contours, viewshed.

Method 3 — Hillshading (Shaded Relief)

Simulates light and shadow on terrain as if illuminated by a light source (conventionally from NW at 315° azimuth, 45° altitude). Bright areas face light; dark areas are in shadow. Creates immediate 3D visual impression without reading contour values.

Method	Data Type	Best Use	Limitation
Contour lines	Vector lines	Topographic maps, engineering design	Hard to interpret for non-experts; no cell-based analysis
DEM	Raster grid	Terrain analysis, watershed, flood modelling	Resolution limits fine-detail capture
Hillshade	Raster (derived)	Visual terrain display, atlas maps	Does not show actual elevation values
Hypsometric tint	Raster classified	General reference maps, school atlases	Colours vary between publishers — not standardised

TIER 1 — APPEARED IN BOTH PAST PAPERS — GUARANTEED

Q5. Explain how Qualitative and Quantitative data can be represented on maps.

Qualitative Data Mapping (Nominal — no order, no magnitude)

Technique	Description	Example
Nominal colour (hue)	Different colours for different categories — no colour implies more or less	Land use: green = forest, blue = water, red = urban, yellow = agriculture
Shape symbols	Different point symbols for different feature types	Circle = hospital, triangle = school, star = police station
Pattern fill	Different textures or hatching patterns for area categories	Geology: dots for sandstone, diagonal lines for limestone

Quantitative Data Mapping (Numeric — has magnitude)

Technique	Description	Example
Choropleth map	Areas shaded in graduated colour — MUST use normalised data (rate/density)	Population density per km² — light to dark colour ramp
Proportional symbols	Point symbols scaled proportional to the data value	Circles sized by city population
Dot density map	Each dot = fixed count of occurrences distributed within area	1 dot = 1000 people — dense dots = high population
Isoline map	Lines connecting points of equal value — like contours for any variable	Isohyet = equal rainfall; Isotherm = equal temperature
Flow map	Lines of varying width showing movement between locations	Migration flows; trade routes between countries

> ★ **EXAM TIP: NEVER use a choropleth map for raw counts (e.g. total population) — always normalise by area (density) or total (percentage). Using raw counts on a choropleth is a fundamental cartographic error.**

TIER 1 — HIGH PROBABILITY | appeared in Paper 1

Q6. Write a short note on Map Dissemination. Explain Cartography and the role of a Cartographer.

Cartography

Cartography is the science, art, and technology of making maps. It involves representing geographic information visually in a way that is accurate, clear, and meaningful to the intended audience — from data collection through to final map publication and distribution.

Map Dissemination

Map dissemination is the process of distributing and sharing finished maps with intended audiences. In the GIS era, maps can be disseminated through multiple channels.

Method	Description	Example
Print / Hard copy	Traditional paper maps printed and physically distributed	Topographic maps, atlas books, survey maps
PDF / Digital file	Static digital maps shared via email or websites	Research papers, government reports
Web GIS / Web maps	Interactive maps hosted online — zoom, pan, query, toggle layers	Google Maps, ArcGIS Online, OpenStreetMap
Mobile applications	Maps optimised for GPS-enabled smartphones	Navigation apps: Google Maps, Waze
Geospatial portals	Online repositories for public spatial data sharing	ISRO Bhuvan, USGS Earth Explorer, Copernicus Hub

TIER 1 — HIGH PROBABILITY | appeared in Paper 1

Q7. How to distinguish between three temporal cartographic techniques? Explain colour in map design.

Three Temporal Cartographic Techniques

Technique	Description	How Time is Shown	Example
Sequential Maps	Multiple separate maps of same area each showing state at a different time — placed side-by-side	Side-by-side static maps labelled with time period	Mumbai land use in 1980, 1990, 2000, 2020
Change Maps	Single map showing difference or change between two time periods	Colours/symbols show: added, removed, unchanged	Deforestation: green = existing

Technique	Description	How Time is Shown	Example
			forest, red = cleared since 2010
Animated Maps	Map updates dynamically as time progresses — time slider or automatic animation	Time slider or animation frames	Animated hurricane track; urban growth animation 1900-2020

Importance of Colour in Map Design

Colour Principle	Description	Example
Nominal — different hues	Distinct colours separate categories — no colour implies more or less	Land cover: green, blue, red, yellow — each = one category
Sequential — light to dark	Single-hue ramp for ordered/quantitative data — darker = more	Choropleth: light yellow → dark red for increasing density
Diverging scheme	Two contrasting hues meeting at neutral midpoint — for data with meaningful centre	Temperature anomaly: blue = below avg, white = avg, red = above avg
Colour blindness	Avoid red/green combinations — use blue/orange or blue/yellow instead	Critical for maps used by general public
Background colour	Light neutral background ensures data colours stand out clearly	Light grey ocean so coloured land features stand out

MASTER CHEAT SHEET — ALL 5 UNITS

Key Definitions — Quick Revision Before Exam

Term	One-Line Definition
GIS	Computer system to capture, store, analyse, and present georeferenced data
Spatial data	Data with a location component — coordinates + descriptive attributes
Topology	Mathematical rules defining spatial relationships: connectivity, adjacency, containment
Tessellation	Dividing geographic space into non-overlapping, space-filling tiles
TIN	3D terrain surface from irregularly spaced elevation points via Delaunay triangulation
Geographic field	Attribute that has a continuous value at every location in a study area
Vector model	Represents features as precise points, lines, and polygons with coordinates
Raster model	Represents space as a regular grid of equal cells each holding one attribute value
Relational model	Organises data into tables (relations) of rows (tuples) and columns (attributes)
SDI	Framework of policies, standards, and data enabling spatial data sharing between organisations
Spatial database	DBMS extended with spatial data types and spatial query operations (e.g. PostGIS)
Metadata	Data about data — source, accuracy, coordinate system, date of a spatial dataset
Map projection	Mathematical method transforming Earth's 3D curved surface onto a flat 2D map
Geoid	Irregular Earth model based on gravity equipotential — corresponds to mean sea level
Ellipsoid	Smooth mathematical approximation of Earth's shape used for coordinates (WGS84)
GPS	Satellite navigation system using trilateration from 4+ satellites for position determination
Interpolation	Estimating values at unsampled locations from known surrounding sample points
Data quality	How accurately and completely spatial data represents the real world
Buffer analysis	GIS operation creating a distance zone around a spatial feature
Overlay analysis	Combining two or more spatial layers to derive new geographic information
Network analysis	Analytical operations on connected linear networks — routing, flow, accessibility
Neighbourhood function	Analysis of each location based on characteristics of its surrounding area

Term	One-Line Definition
Error propagation	Accumulation and magnification of input data errors through spatial operations
DEM	Raster grid where each cell stores terrain elevation in metres
Hillshading	Simulated light and shadow on terrain based on sun direction — creates 3D effect
Cartography	Science, art and technology of making maps
Visual variable	Graphic property of map symbols that can be varied: size, shape, hue, value, orientation, texture
Bertin's variables	Six graphic variables (SSCOOT) for encoding geographic information on maps
Choropleth map	Areas shaded by graduated colour to show a normalised statistical variable
Map cosmetics	Non-data map elements: title, legend, scale bar, north arrow, graticule, border, source
Map dissemination	Process of distributing finished maps to audiences via print, web, or mobile channels
Temporal mapping	Representing geographic change over time using sequential, change, or animated maps